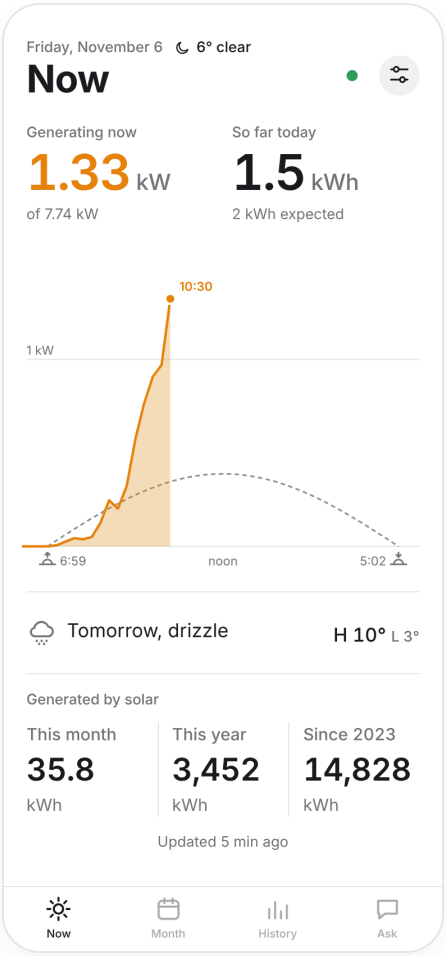
# Solar

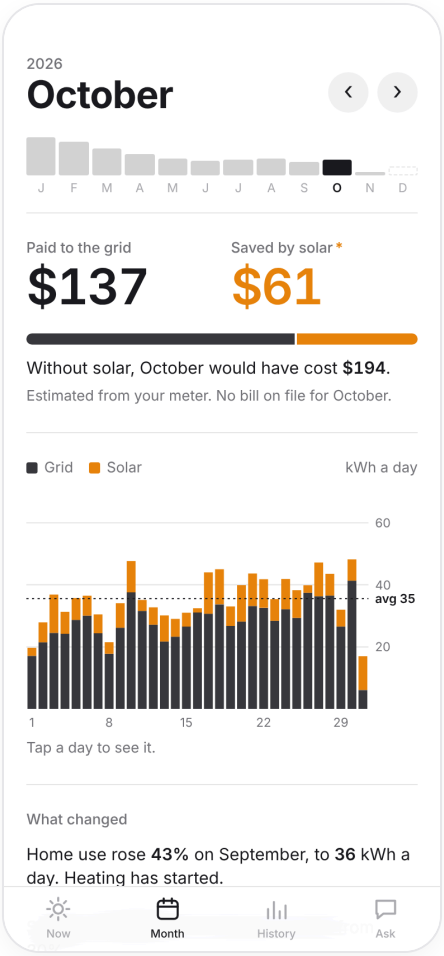
A private app for one home: what the panels make, what power costs, and what solar saves.

Solar joins two sources that never talk to each other: the **SolarEdge inverter** on the house (17 panels, 7.7 kW) and **Toronto Hydro's** meter readings and bills. It works out Ontario's net metering rules the way the bills do, so the savings are real dollars, checked against every bill on file. It opens to four tabs, Now, Month, History and **Ask**, where you can ask about your energy by typing or by voice, and it runs from the iPhone Home Screen.



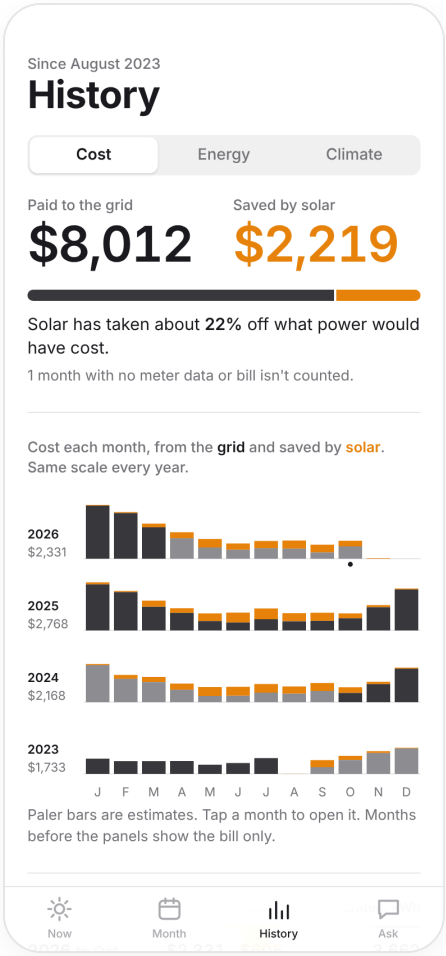

## Now

What the panels are making, today so far against the expected curve, sunrise and sunset, weather and the totals since 2023. A dot shows the system's health.




## Month

What you paid and what solar saved, with what the month would have cost without it. Every day split into grid and solar. Tap a day or open the bill.




## History

Every month since 2023, one row per year on one scale. Switch to Energy for what the panels made, or Climate for heating and cooling.

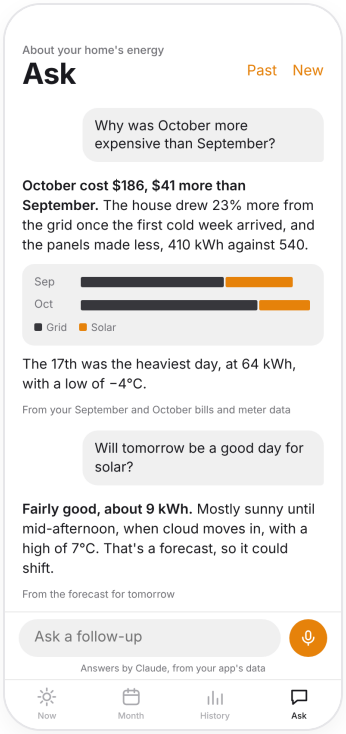
### AT A GLANCE (SAMPLE FIGURES)

**14,828**
kWh generated since August 2023

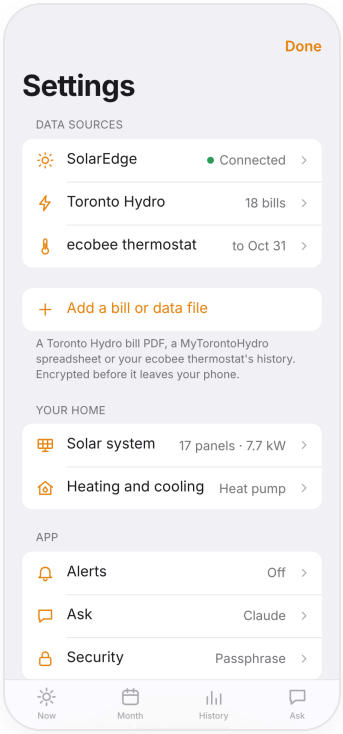
**$2,219**
saved by solar, about 22% off the power bill

**Every**
Toronto Hydro bill checked line by line against the meter

**1 hr**
between updates, with alerts when something's wrong




**Ask**
Questions in plain words, answered from the app's own data, with a small chart when it helps.




**Settings**
Every source and setting on one short page, each with its status. Tap a row for its details.

## Ask

Type a question or tap the mic and say it. Claude answers in a few plain sentences with the key number first, from your months, bills, the last 120 days and the live forecast, and names the data it used. Spoken questions are answered aloud. Conversations are kept on the phone under Past.

## Live from the inverter

Power now, today's output and the expected curve, which levels off at the inverter's 6 kW limit. At night a small moon shows the real phase; tap it for its name.

## Costs that match the bills

Credits for power sent back are valued at the time-of-use price when it went out, delivery is charged on net use, and HST and the Ontario rebate are applied as Toronto Hydro does. Months without a bill are estimated from the meter, usually within a few percent.

## Bills, meter and thermostat data

Add bill PDFs, MyTorontoHydro spreadsheets or the ecobee's history from Settings, several at once or in a zip, encrypted on the phone first. A bill is only added if it adds up. The ecobee file adds heating to Month and a Climate view to History.

## Alerts

Phone notifications when production stops on a sunny day or the inverter reports a fault, when SolarEdge stops sending data, when a file is added, a summary on the 1st of each month, and in winter, days of unusually heavy use with the temperature.

## Private by design

All data is encrypted with a passphrase before it's published. Unlock with the passphrase, or with Face ID on a trusted phone. Serial numbers and tokens live only inside the encrypted data.

## How it runs

A scheduled job on GitHub fetches SolarEdge and the weather about every hour, reads any new files, rebuilds the site on GitHub Pages and sends alerts. There's no server to maintain.

## Your system

Panels, optimizers, inverter, rapid shutdown switch, meter and installer, with serial numbers for warranty or service, and what to do in an emergency.

## Good to know

Meter data comes from exported files, so costs trail by however long it's been since the last one. SolarEdge's API has no per-panel data. An answer in Ask is lost if you leave Solar mid-answer; it's asked again when you come back.

SOLAR

# Technical setup

## SOURCES AND LIBRARIES

| PIECE | VERSION | WHAT IT DOES | LOADED FROM | LICENCE |
|---|---|---|---|---|
| **App code** | App 1.61 | The four tabs, charts, net metering maths, encryption and Ask (`app.js`, `lock.js`, `style.css`, `sw.js`). No framework; charts and icons are SVG drawn for the app, text in the system font | GitHub Pages, this repo's `web/` | Written for the app |
| **Build job** | Python 3 | Fetches data, reads bills and meter files, models costs, encrypts, sends alerts (`scripts/build.py`, `money.py`, `parse_bill.py`) | GitHub Actions | Written for the app |
| **Transformers.js** | 3.5.1 | Runs the speech model in the browser, with ONNX Runtime Web inside | cdn.jsdelivr.net | Apache-2.0 |
| **Whisper tiny.en** | Xenova, 8-bit | Turns a spoken question into text on the phone (about 40 MB, cached after first use) | huggingface.co | MIT |
| **Python libraries** | latest | cryptography (AES-GCM), openpyxl (meter spreadsheets), pywebpush (notifications) | PyPI | Apache-2.0, MIT, MPL-2.0 |
| **Poppler pdftotext** | Ubuntu package | Reads the text of bill PDFs | apt | GPL-2.0 |

## INTEGRATIONS

| SERVICE | USED FOR | DETAILS |
|---|---|---|
| **SolarEdge Monitoring API v2** | Production, power now, inverter status and alerts | `monitoringapi.solaredge.com/v2`, OAuth through `connect.solaredge.com`; the refresh token rotates and is saved back as a GitHub secret. Free tier, 2,000 calls a month |
| **ecobee** | Heating history | No API for new developers; the Download data CSV is added from Settings and kept encrypted in `data/ecobee/` |
| **Toronto Hydro** | Bills and hourly meter readings | No API. Bill PDFs and MyTorontoHydro spreadsheets are added from Settings; kept encrypted, the PDF isn't kept |
| **Open-Meteo** | Weather now, forecast, past weather | `api.open-meteo.com` and `archive-api.open-meteo.com`, no key. Called by the build, the Now screen, and Ask for each question |
| **Anthropic Claude API** | Answers in Ask | `api.anthropic.com/v1/messages`, model `claude-haiku-5-5`, called from the phone. Sends the question, recent conversation and a summary of months, bills, 120 days and the forecast. Key in the `ANTHROPIC_API_KEY` secret, carried inside the encrypted data |
| **GitHub** | Hosting, the hourly job, uploads | Pages serves the site, Actions runs the build hourly, and files from the phone arrive encrypted in `data/inbox/` via the Git Data API (`UPLOAD_TOKEN`) |
| **Web Push (Apple)** | Alerts on the phone | VAPID-signed pushes from the build; subscriptions kept encrypted in `data/push.enc` |
| **Browser features** | Unlock, voice, offline | Web Crypto (PBKDF2-SHA256, AES-GCM), passkeys for Face ID unlock, microphone and Web Audio for recording, speech synthesis to read answers, a service worker, and local storage for Ask conversations |

## What runs where

On the phone: unlocking and decrypting, every screen and chart, transcribing spoken questions, reading answers aloud, and the calls to Claude and Open-Meteo. Files you add are encrypted before they leave.

On GitHub, about every hour: the build fetches SolarEdge and the weather, reads new files, rebuilds the site and sends alerts. The site's data is published encrypted with a key from the passphrase (PBKDF2-SHA256, 600,000 rounds); serial numbers and tokens exist only inside it.

This document: `docs/overview/`. Make sample data with `fake_data.py`, take screenshots with `shoot.js`, print with `pdf.js`.